acoem

Fusion

Smart Sound & Vibration Anaslyser

Acoem's Fusion is the most versatile Class 1 sound level meter on the market, designed for acoustic consultants, experts, and technicians alike. From handheld measurements and building acoustics to environmental noise monitoring and long-term studies, Fusion adapts to any challenge with precision and accuracy.

Discover the Fusion difference:
Unrivaled versatility Intuitive operation for simple and advanced tasks **Seamless** integration with Cadence cloud platform

Explore the datasheet to learn how Fusion can revolutionize your acoustic workflow.

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Fusion

Certified to the highest standards (IEC 61672), FUSION delivers unparalleled versatility and performance in a single, easy-to-use instrument. Capture acoustic, vibration, and audio data simultaneously, even with wireless sensors. Boost your productivity with seamless remote control and powerful processing software.

General Overview

1. Class 1 microphone
2. Integrated preamplifier
3. Colour display
4. Keyboard
5. Windscreen
6. Removable hand grip
7. Rail fastening
8. Mini HDMI
9. DC 8-28V power supply input
10. RJ45 Network
11. External microphone preamplifier input & analogue output
12. Mini USB
13. SIM card slot
14. RS232 input
15. TTL input/output
16. SD card slot

Deliverable & Accessories

The standard package (FSN1001000) of FUSION includes the following items

Handle
Profile for tripod mounting

Windscreen
Windscreen claw

Metrological Documentation

SD card 2GB

AC power supply
USB Cable

Cadence & Fusion

Fusion is versatile and can be used as dedicated noise monitoring system designed for seamless integration into your existing setup. To further streamline your monitoring projects and reduce operational costs, Acoem offers Cadence, a suite of web services that simplifies deployment, management, and maintenance.

Cadence

User-friendly platform puts the power of acoustic monitoring in everyone's hands.

Data Security and Control

Cadence secures your acoustic data and provides real-time information on sound levels, system alerts, and triggers.

Optimized Communication

Accelerate decision-making by sharing relevant information with the right people. Create public consultation sites and configure real-time alerts.

Integrated Artificial Intelligence

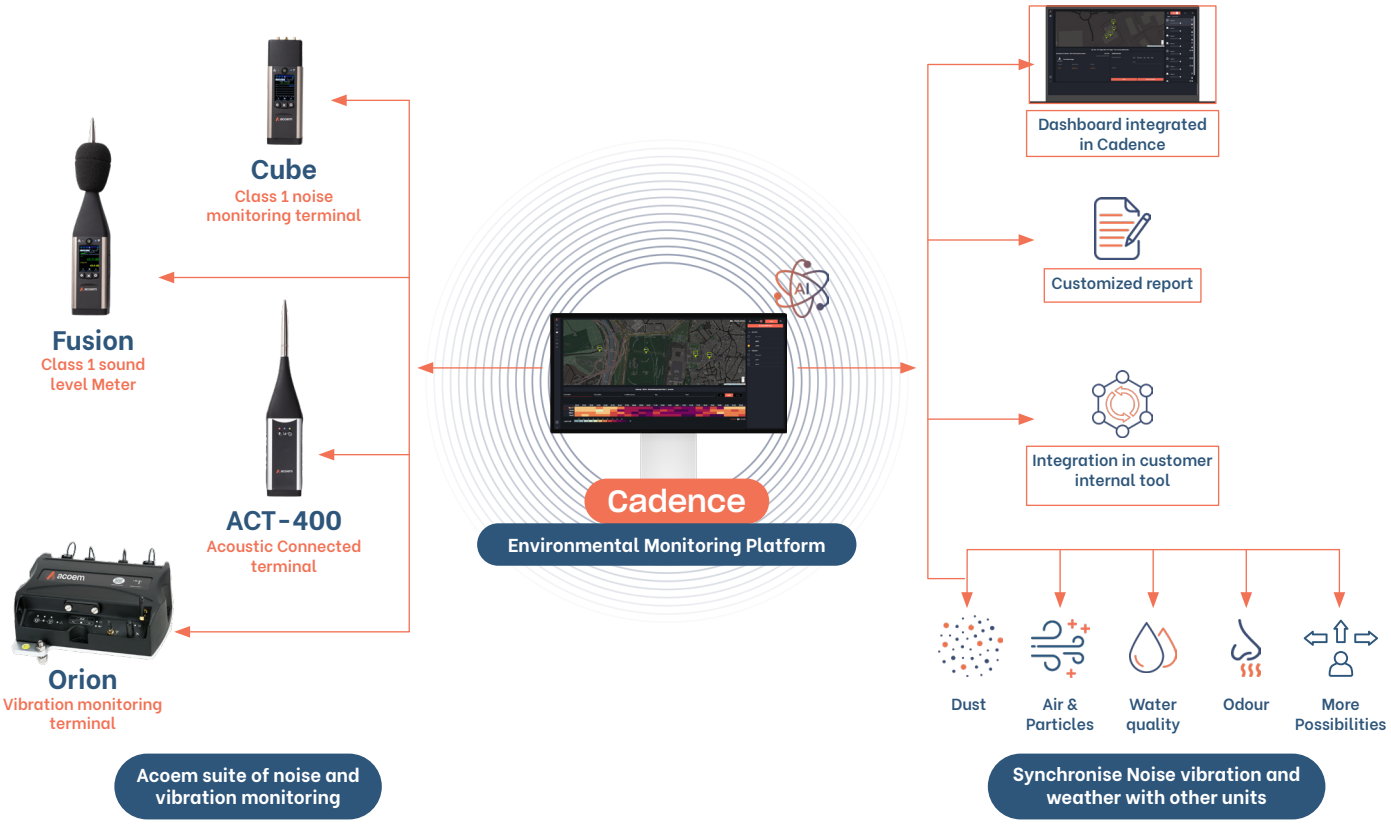
Automatically identify problematic noise sources with Cadence's pre-trained AI. Save time and simplify acoustic analysis without specific expertise.

Seamless Integration

Integrate Cadence with your existing systems via documented APIs. Exchange real-time and aggregated data with your other tools for optimized monitoring.

Advanced Connectivity

Transform your sound level meters into connected objects. Remotely manage your Acoem devices: configuration, updates, metrological monitoring, and scheduling.



Technical Specifications (serial number > 14000)

IEC class
IEC 61672-1 ed. 2.0 (2013) (0° & 90° reference direction)
IEC 61620 (1995) NF EN 61260/A1 (2002)
Sound Level Meter, Integrating Sound Level Meter with storage, group X.

Type approval (extract)
France: N° LNE-29639 rév. 3 du 07 juin 2021
Switzerland: METAS: CH-A3-2111 0-00
Germany: PTB: DE-16-M-PTB-0008, Revision 2
Portugal: NESMC-11
Spain: CEM: 210491003/M2

Dynamic range
21-139 dB (A, B), 26-139 dB (C), 31-137 dB (Z),
One single range for a rated sensitivity of 40 mV/Pa

Linear operating range for A weighting (5 frequencies)
31,5 Hz : 26-98 dB
1 kHz : 23-138 dB
4 kHz : 23-138 dB
8 kHz : 23-134 dB
12,5 kHz : 23-130 dB

Dynamic range Peak
60-140 dBC, 1 single range

Time weightings
Slow, Fast, Impulse, Peak

Frequency weightings
X=A, B, C, Z; Y=S, F, I for LXeq & LXY X=A; Y=S, F, I for LXYTd
X=C, Z for LXpk

Instantaneous broadb& values stored

	LXY				LXYeq				LXYTd				LXYMinMax			
	A	B	C	Z	A	B	C	Z	A	A	B	C	Z
F	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
S	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
I	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Pk			✓	✓									

LnsT (sliding Ln) | LAeqsT (sliding LAeq)
LAexPT (exposure level)

Instantaneous weather data stored

Fe[kHz]	48	32	16	8
Possible bit rate [kbps]	32	32	8	8
	96	96	32	32
	192	192	96	96

Wind speed [m/s] Wind direction [°] Rain intensity [mm/h]
Barometric pressure [hPa] Air temperature [°C] Humidity [%HR]

Noise logging period T
Mini 20ms - maxi 3600s, 5 ms steps
Short logging period: mini 20ms - max standard T, 5 ms steps. Short logging period applicable during events Short logging period must be a divisor of T

Weather logging period
Weather logging period is a multiple of T with a minimum of 1 second

Spectral analysis
Parallel measurement & storage of Leq & LY (Y=F, S, I)

Filters
1/1 (8Hz-16kHz) et 1/3 (6.3Hz-20kHz)

Statistics
7 selectable Ln in parallel from L1 to L99, 1 dB class Samples for calculation: T if Leq or 20 ms if LXY, 0.1 dB resolution.

Back erase
0, 5s or 10s, SLM mode only

Input high pass filter
0,3 Hz / 10 Hz

Reference directions
0° on internal input
0° & 90°, selectable built-in correction on external input (with a DMK01 external microphone)

Reference point for microphone
Centre of the protection grid (with or without nose cone).

Calibration
Reference level: 94 dB
Acoustic calibration detection

Starting point for linearity tests
Reference level, i.e. 94 dB.

Data storage modes
SLM (hand-held sound level meter) & LOG (logging sound level meter)

Signals recording
Audio signal type
allows selecting either RAW (uncompressed data for dBTrait post-processing) or MP3 (format MPEG-1/2 Audio Layer 3 compressed data). allows selecting audio signal sampling frequency; Possible choices:
For RAW : 51.2kHz, 25.6kHz, 12.8kHz, 6.4kHz,3.2kHz,1.6kHz)
For MP3 : 48kHz, 32kHz, 16kHz, 8kHz.
MP3 bit rate [kbps] allows selecting MP3 bit rate ; possible

Triggers (5 different per event)
Settings for pre-trigger, post-trigger, minimum time, end time

Types: on instant acoustic and weather values (except wind direction), instant spectral values, TTL input.

Manual markers
On the instrument: 1 code “code 1”
On web interface: 5 codes: “codes 1 to 5”

Timers
Immediate, differed, daily periodic Audio: periodic

Typical background noise
Microphone (thermal noise): 14.5 dBA, 15.0 dBC, 15.3 dBZ

	[dB]	LA	LC	LZ
Acoustic	Typical	16,1	16,8	20,2
	Maximum	20,0	21,0	24,0
lectric	Typical	11,0	12,5	18,5
	Maximum	16,0	16,0	21,0

Vibration
Signal: Metrological, Fs = 12,800 Hz
Pre-trigger = 0 sec
1 (Z) or 3-axis(X, Y & Z)

Audio recording triggers
Simultaneously with events & manual
(using FUSION integrated key & web interface
for remote control)

Events (automatic coding)
5 user-definable event: codes 6 to 10 24
user-definable time periods

Preamplifier
Integrated, not removable
External type PRE22 (included in DMK01) on external input
(standard 10 m lemo extension cable)

Integrated keys
4 silent keys: on/stand-by/off & 3 multi-functions keys

Status indicators
LED red (overload)
LED blue (Wi-Fi connection)
LED green (power ON, blinking on on-going
measurement, charge ON)

Display
High contrast colour screen 38*50mm resolution 320*240 pixels 3 sets of colours (day, contrast, night) Display rate: 0.1s, Display resolution: 0.1dB

USB connection
Type 2.0; mass storage mode, charge on USB

Ethernet connection
Connector RJ45, Speed: 100 MB/s DHCP(automatic) and Manual modes available

Wi-Fi Connection: (Antennas included)
IEEE 801.11b, g
Point-to-point connection & infrastructure mode

Cellular network connection (Antennas included)
Embedded modem 3.5G compatible with 4-band GSM/GPRS/ EDGE & 3-band UMTS/HSDPA

Data connectivity
Integrated Network protected http server for web interface
Integrated FTP server for data access

SMS alarms
On event:
SMS text with CUBE serial #, location, date and time, user defined text, IP address:http port

On low battery (10%):
SMS text with CUBE serial #, location, date and time, % remaining battery

On movement:
SMS text with CUBE serial #, location, date and time, GPS coordinates, distance from previous location, IP address:http port (the alarm trigs if CUBE has moved more than the user defined distance)

Automatic SMS actions
Sending “IP” by SMS to instrument makes it reply by sending an SMS with instrument serial #, location, date & time, IP:port address & automatically sends a new SMS at every new IP address in case of floating IP

Actions on SMS sent to the instrument
On SMS sent “IP”, the instrument replies by sending an SMS with the instrument serial #, location, date & time, IP:port address On SMS sent “stop”, the instrument stops replying new SMS if IP has changed On SMS “reboot”, the instrument reboots to establish a new connection & replies with an SMS with instrument serial #, location, date & time, IP :port address

Web interface refresh rate webpages
Standard: twice per second Mobile: once per second

Analogue output
Audio output A, B, C or Z (+/-10Vpp R=2000hms)
Adjustable gain: 0, 10, 20, 30, 40, 50 dB

Electrical check
Programmable periodicity: 1, 2 or 4 times per day (0h,0h-12h, 0h, 6h, 12h, 18h)
3 pre-set frequencies (1000 Hz, 2000 Hz, & 4000 Hz) & 2 user-defined frequencies (between 10 Hz & 20 kHz)
2 user-defined excitation levels, maximum level 5 V (100%)

External microphone input
For DMK01, PRE22 (R = 560k0ms / 22Vpp (+/- 11V)

TTL output
R = 100 0hms / 0 / 5V

TTL input
R = 100 k0hms / 0...1V = «0» 1.8...5V =»

Typical power consumption
Without communication (screen switch off): < 1200 mW
+ Wi-Fi & screen switch on: < 1800 mW
+ Modem: <3800 mW

Operating lifetime
20 hours with Wi-Fi connection (during 10% of measurement time) (for temperatures ranging from 10°C to 50°C, in LOG mode with IT = 1 s, fine IT 100 ms, 1/3 octave & audio recording on threshold during 10% of the measurement time)

External power supply
DC 8 to 28 V on charge input
DC 5 V on USB input (slow charge)

Memory
SD, SDHC or SDXC card, 2 GB or higher (2GB standard delivery) for measured data & signals. Minimum recommended requirement: ≥ class 10. Please note only SD cards provided by Acoem should be used. Acoem cannot be held responsible for data loss if the SD card used is not delivered by Acoem.

Measured data stored on the SD card every 10 seconds. Non-volatile memory for configurations, system log (500), calibration data (500) & electrical checks (500)TTL input R = 100 kOhms / 0...1V = «0» 1.8...5V =»1»

Battery
Type lithium polymer Voltage 3.7V Capacity 6750 mAh Non-removable, charging time approximately 3 hours

Clock
GPS PPS, error < 50 milliseconds Internal clock, error < 0.5 s/24 hours

Localization
Automatic with integrated GPS
Information stored with measurement campaigns

Warm-up time
From power off: < 25 seconds

Operating temperature:
-10°C to +50°C

Humidity
IEC 60068-2-78: damp heat: 90% HR
(non condensing at 40°C).

Electromagnetic compatibility
According to Directive 2004/108/EC
NF EN 61000-6-1 NF EN 61000-6-2 NF EN 61000-6-3
NF EN 61000-6-4 (2001)
ETSI EN 300 328 V1.5.1 (2004)

Protection
IP40 in standard use
IP42 if the instrument is used in vertical position without connectors cover
IP55 if the instrument is used in vertical position with connectors cover

Influence of vibration
Use with no outdoor microphone:
For mechanical vibration of an acceleration level of 1 m/s² perpendicular to the microphone diaphragm, at frequencies 31.5 Hz, 63 Hz, 125
Hz, 250 Hz, 500 Hz, 630 Hz, 800 Hz & 1000 Hz:
the lower limit of the linear operating domain for A-weighting becomes 80 dB.

For mechanical vibration of an acceleration level of 1 m/s² parallel to the microphone diaphragm, at frequencies microphone diaphragm, at frequencies 31.5 Hz, 63 Hz, 125 Hz, 250 Hz, 500 Hz, 630 Hz, 800 Hz & 1000 Hz: the lower limit of the linear operating domain for A-weighting becomes 60 dB.

Use with outdoor microphone unit DMK01:
For mechanical vibration of an acceleration level of 1 m/s² perpendicular to the microphone diaphragm, at frequencies microphone diaphragm, at frequencies 31.5 Hz, 63 Hz, 125Hz, 250 Hz, 500 Hz, 630 Hz, 800 Hz & 1000 Hz:
the lower limit of the linear operating domain for A-weighting becomes 75 dB.

Weight & dimensions
775 g
H x L x P: 300 x70 x 52 mm

Building Acoustics Module (Opt.FSN2009000)
CALCULATION OF REVERBERATION TIMES
Fine IT 20 ms for decay analysis Simultaneous calculation of T20 & T30

Automatic detection of interrupted or pulsed noise sources Schroeder integration for pulsed sources Estimate by least squares approximation

Calculation of quality indicators (Iso 3382)

Name	Quality indicator	Description, default values
N	Background noise level too high*	Low dynamic range (between 41 & 45 dB for T30; between 31 & 35 dB for T20)
D	Calculation impossible*	Insufficient dynamic range (< 41 dB for T30;< 31 dB for T20)
<	Reverberation time too low*	Tr < 0.24 seconds (scaled by logging period = 20 ms)
ξξ	Non-linearity*	Non-linearity parameter >1%
C	Curvature*	C > 10% or C < 0; see [1] appendix B.3 Difference
L	Linearity of the sound source linearity	Difference between adjacent 1/1 or 1/3 octave bands > 6 dB

Frequency-based analysis

1/1 or 1/3 octave, 50 to 5000 Hz

Levels L1, L2, Li (Emission, Reception, Impact Noise)
Calculation of the mean spectrum LZeq over the specific coding duration, detected automatically (source on duration).

Background Noise Level (Db)
Calculation of the mean spectrum over the entire Measurement duration

ISO 3382-2 standard indicator
Invalid indicators displayed on the Tr spectrum & stated on decay

Audio Comments
Used to store a voice comment, with the same sampling frequency as for the measurement

Pc Software
dB Inside

Integration Times (It)
1 second; 20 milliseconds

Maximum Averaging Time For Spectra L1, L2, Lb & Li
120 seconds

Maximum Measurement Time For Equipment Noise
600 seconds

Simultaneous Audio Recording
Sampling frequency: 51.2 kHz, 25.6 kHz, 12.8 kHz, 6.4 kHz, 3.2 kHz, 1.6 kHz

Equipment Noise Levels
Selection of the maximum level for one of the following parameters: LXYMax where X = A, C or Z & Y = F, S or I

Optional Accessories
Weather Station and anemometer:

	WXT532	ANEMO	WXT536
Wind speed	✓	✓	✓
Wind direction	✓	✓	✓
Air temperature		✓	✓
Relative humidity			✓
Rain intensity			✓
Barometric pressure			✓

Wireless vibration sensor 3-axis (X, Y, Z) 80g, Weight 373 g, Dimension Ø42 x H116 mm, 8h battery life.
Weatherproof external charger IP67 (10m cable)
Connecting these accessories has no Influence on measurements.

DMK01
(External Microphone)

Kit weatherproof -
(Standalone)

CAL31 Class1 Calibrator

DSC01 Outdoor case

TM01 Tapping Machine

Omnidirectional Sources
- LS01 LS02 LS03 LS04

Anemometer

WTX Weather Station